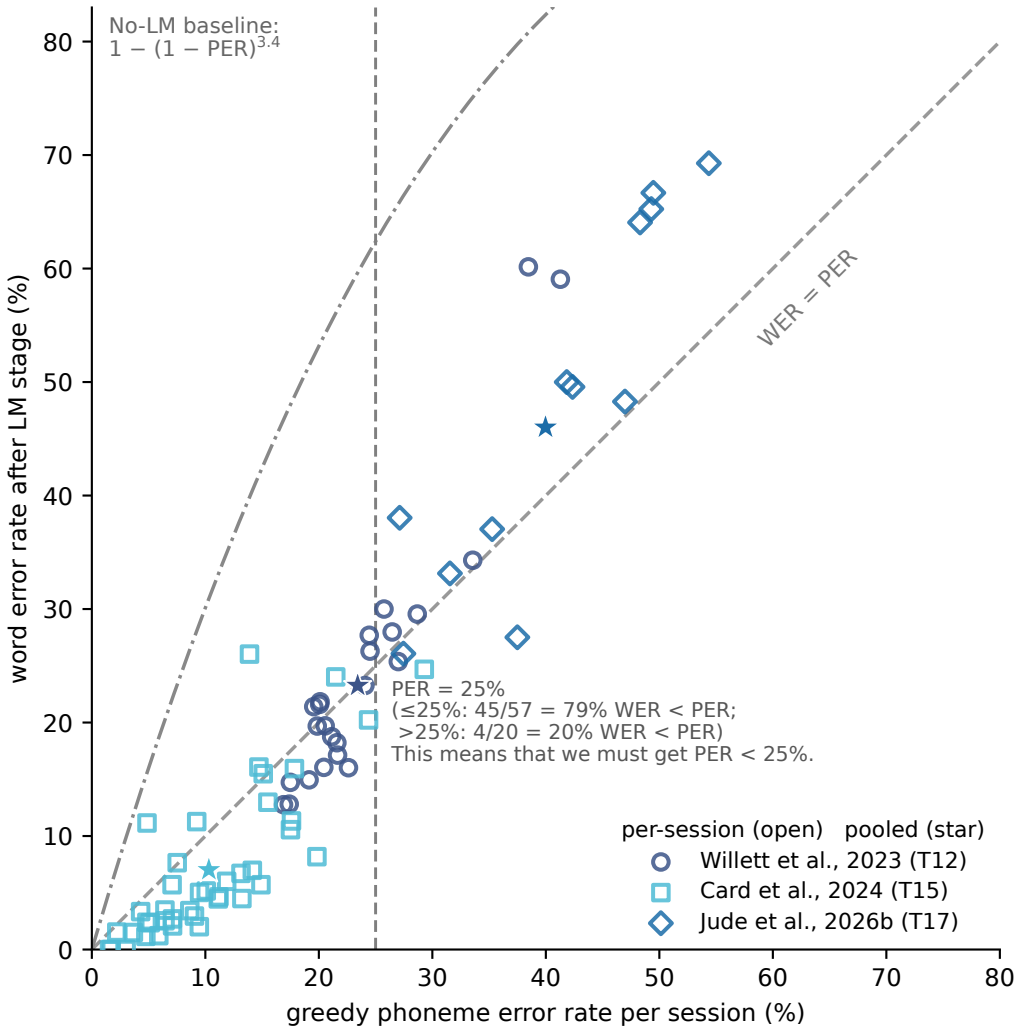
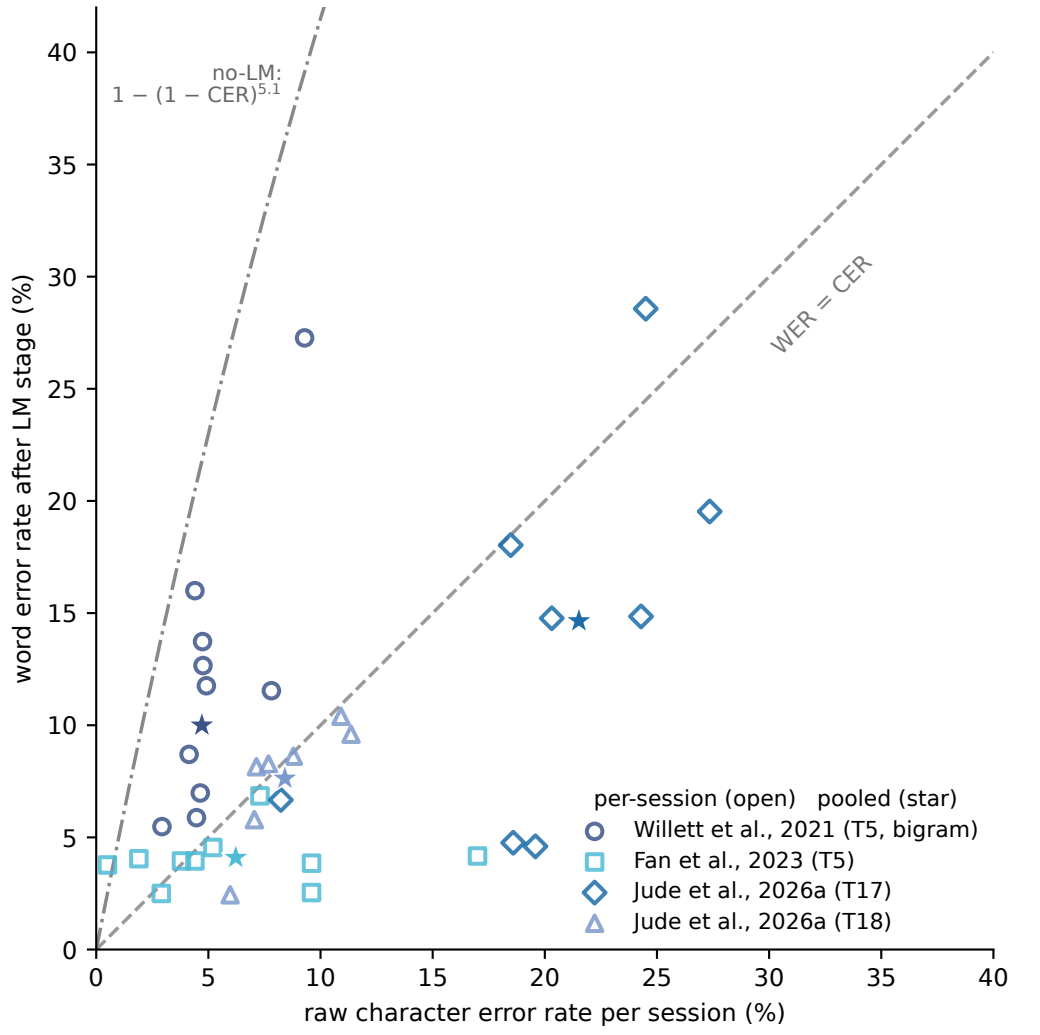
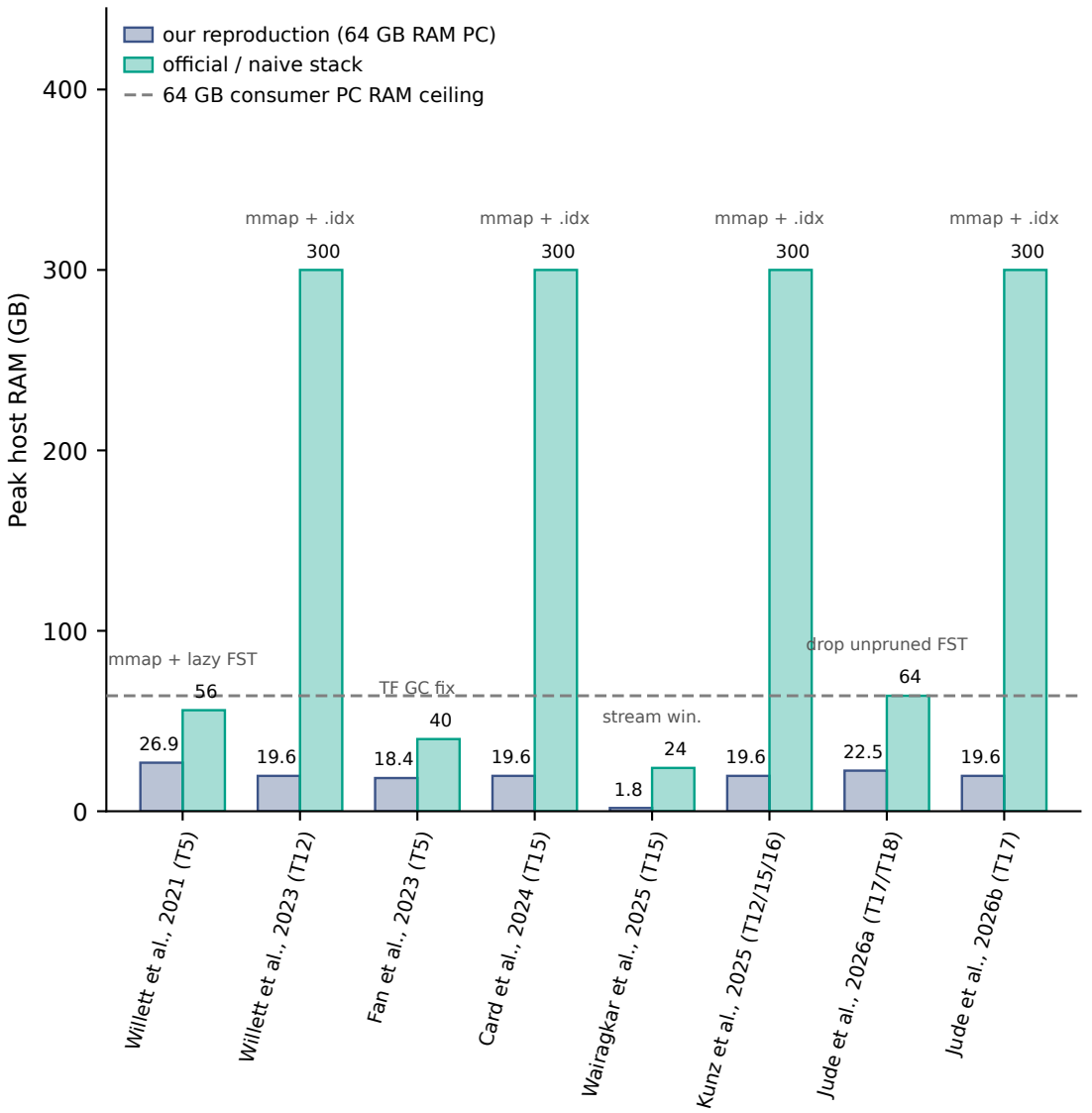
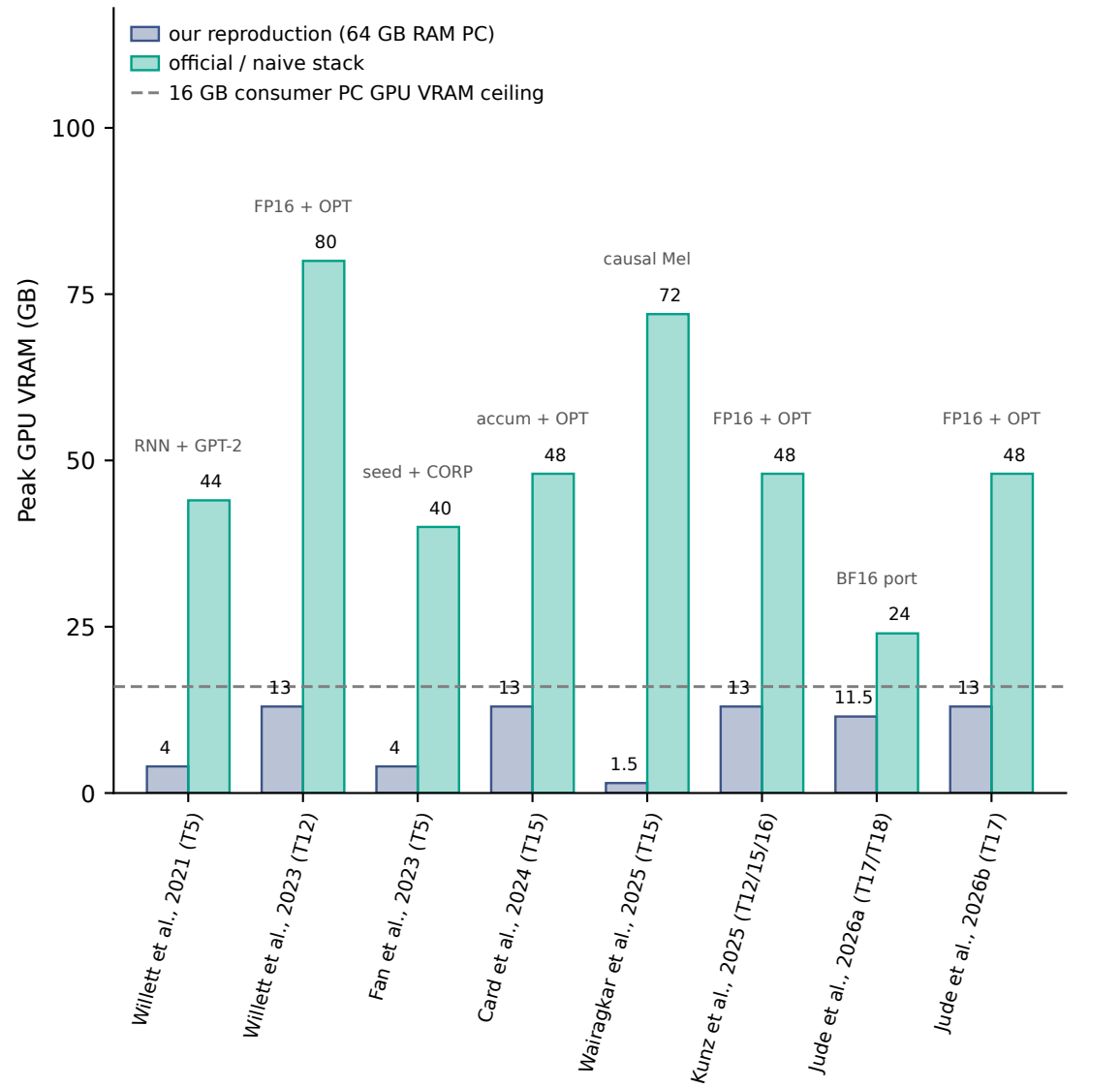

(a) Session-level PER vs. post-LM WER
No-LM baseline:
1 − (1 − PER)^3.4
WER = PER
PER = 25%
(≤25%: 45/57 = 79% WER < PER;
>25%: 4/20 = 20% WER < PER)
This means that we must get PER < 25%.
per-session (open) pooled (star)
Willett et al., 2023 (T12)
Card et al., 2024 (T15)
Jude et al., 2026b (T17)
word error rate after LM stage (%)
greedy phoneme error rate per session (%)


(b) Session-level CER vs. post-LM WER
no-LM:
1 − (1 − CER)^5.1
WER = CER
per-session (open) pooled (star)
Willett et al., 2021 (T5, bigram)
Fan et al., 2023 (T5)
Jude et al., 2026a (T17)
Jude et al., 2026a (T18)
word error rate after LM stage (%)
raw character error rate per session (%)


(c) Host RAM across all 8 studies: our 64 GB PC vs. official stacks
our reproduction (64 GB RAM PC)
official / naive stack
64 GB consumer PC RAM ceiling
Peak host RAM (GB)
mmap + lazy FST
TF GC fix
stream win.
drop unpruned FST
mmap + .idx
26.9
56
19.6
300
18.4
40
19.6
300
1.8
24
19.6
300
22.5
64
19.6
300
Willett et al., 2021 (T5)
Willett et al., 2023 (T12)
Fan et al., 2023 (T5)
Card et al., 2024 (T15)
Wairagkar et al., 2025 (T15)
Kunz et al., 2025 (T12/15/16)
Jude et al., 2026a (T17/T18)
Jude et al., 2026b (T17)


(d) GPU VRAM across all 8 studies: our 16 GB GPU vs. official rigs
our reproduction (64 GB RAM PC)
official / naive stack
16 GB consumer PC GPU VRAM ceiling
Peak GPU VRAM (GB)
RNN + GPT-2
FP16 + OPT
seed + CORP
accum + OPT
causal Mel
FP16 + OPT
BF16 port
FP16 + OPT
4
44
13
80
4
40
13
48
1.5
72
13
48
11.5
24
13
48
Willett et al., 2021 (T5)
Willett et al., 2023 (T12)
Fan et al., 2023 (T5)
Card et al., 2024 (T15)
Wairagkar et al., 2025 (T15)
Kunz et al., 2025 (T12/15/16)
Jude et al., 2026a (T17/T18)
Jude et al., 2026b (T17)